Synopsis

TRAFFIC ACCIDENT SEVERITY PREDICTION

Problem Id: 20

Traffic Accident Severity Prediction

This system predicts accident severity using data like speed, weather, road condition, and vehicle type. Preprocessing cleans dataset and removes irrelevant records. Feature selection keeps most impactful factors like speed and weather condition.

Logistic Regression and SVM (using 3 kernels – Linear for simple separation, Polynomial for curved boundaries, RBF for complex patterns) are used. Models are evaluated using confusion matrix and recall score.

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Section -- G

- **Problem Description:**

Traffic accidents are a serious problem all over the world. They can cause injuries, loss of life, and also financial damage. The main aim of this project is to predict how severe a traffic accident can be by using machine learning. The prediction is based on factors like speed of the vehicle, weather conditions, road condition, and type of vehicle.

This system can help in identifying risky situations in advance. It can be useful for traffic police, emergency services, and even drivers to take better decisions. For example, if the system predicts high accident severity in bad weather or at high speed, some safety measures can be taken.

In real life, this kind of system can help in improving road safety and reducing the impact of accidents. It can also be useful in smart traffic systems and for insurance companies to understand risk better.

- **Steps of Implementation:**

This section should explain the each phase of ML pipeline used.

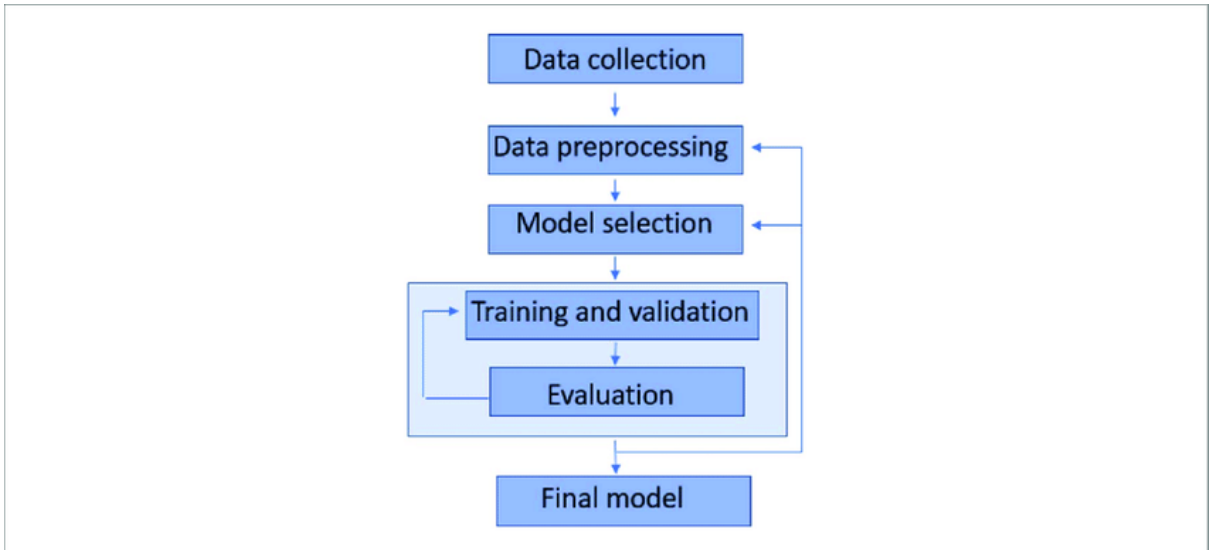

Fig. 1 Sample architecture

Data Collection

Gather dataset from sources

Example: Accident data (speed, weather, road condition)

Data Preprocessing

Clean data

Handle missing values

Encode categorical variables

Normalize data

Feature Selection

Select important features

Example: Speed, Weather

Data Splitting

Divide into:

Training data

Testing data

Helps evaluate model properly

Model Training

Apply algorithms like:

- Logistic Regression
- SVM

Model Evaluation

Check performance using:

Confusion Matrix

Recall, Accuracy

Measures correctness

Prediction / Deployment

Use trained model on new data

Predict accident severity

- **Expected Modules/ Libraries to be used :**
- **NumPy:**
  Used for numerical computations and efficient array operations in machine learning tasks.
- **Pandas:**
  Used for data manipulation, cleaning, and organizing datasets into structured form.
- **Matplotlib:**
  Used for creating basic graphs and visualizations to understand data patterns.
- **Scikit-learn:**
  Used to build and evaluate machine learning models like Logistic Regression and SVM.
- **Expected Evaluation metrics and Plots to be used:**
  - **Confusion Matrix:**
    Shows how many predictions are correct and incorrect in a simple table format.
  - **Recall Score:**
    Tells how well the model can correctly identify severe accidents.
  - **Accuracy:**
    Shows the overall percentage of correct predictions made by the model.
  - **Plots:**
    - Heatmap of confusion matrix to visualize results clearly
    - Feature importance graph to see which factors matter most
    - Model comparison graph to compare performance of different models

These metrics help in understanding how well the model is working, especially for predicting serious accidents.

- **Required topics from the subject:**

•Strings:
In Python, strings are used to handle textual data such as weather conditions, road types, and vehicle categories. They help in data preprocessing, cleaning, and converting categorical values into usable formats.

• Lists & Dictionaries:
Lists are used to store datasets and features, while dictionaries help in mapping categorical values (e.g., weather → numeric labels). These structures make data handling efficient and organized.

• NumPy Arrays:
NumPy is used for handling large numerical datasets efficiently. It provides fast operations for mathematical computations required in machine learning models.

• Pandas DataFrames:
Used for data manipulation, cleaning, filtering, and preprocessing. It helps in reading datasets (CSV files) and preparing data for model training.

• **File Handling:**
Python is used to read and write data files (like CSV). This is important for loading datasets and saving results.

• **Data Visualization:**
It represents data using graphs and charts to easily understand patterns and trends. Tools like Matplotlib and Seaborn are used.

• **Data Preprocessing:**
Includes handling missing values, encoding categorical data, feature selection, and normalization to prepare raw data for models.

• **Machine Learning:**

Includes supervised learning, classification, data preprocessing, feature selection, train-test split, Logistic Regression, SVM with different kernels, model evaluation using confusion matrix and recall, and concepts like overfitting and ML pipeline."

- **Platform Required:**
- Jupyter Notebook
- Python IDLE
- Visual Studio Code

- **Books and Link Sources:**
  1. Scikit-learn documentation
  2.GeeksforGeeks -ML concepts
  3. Kaggle -Dataset repository